

- » Working with a wireless access point
- » Configuring Windows for a wireless network
- » Securing a wireless network

Chapter 2

Wireless Networks

Nearly all modern networks include wireless access. In Book 1, Chapter 2, you learn the basics of how a wireless network works and how to incorporate wireless access in your networking plan. In this chapter, you learn the basics of configuring and securing a wireless network.

Installing a Wireless Access Point

When planning your wireless network, you'll find two distinct devices that provide wireless access:

- » **Wireless firewall routers:** Wireless firewall routers combine the function of a wireless access point with the function of a router and a firewall. This type of device is also sometimes called a *multifunction wireless router* or a *gateway* device and is often supplied by your internet service provider (ISP).
- » **Wireless access point (WAP):** WAPs provide just the wireless access point function. These are sometimes referred to just as *access points* (APs).

For a small office or home network, you'll probably just use a wireless firewall router provided by your ISP. For larger offices, you'll want to use dedicated AP

devices rather than the access point that's built in to the gateway devices provided by your ISP. If your office is large enough, you'll need two or more AP devices, strategically placed throughout the office to provide coverage to every nook and cranny of the space.

The physical setup for a wireless AP is pretty simple: You take it out of the box, mount it in its desired location, and plug it in. You can place the access point on top of a bookcase or shelf, or you can mount it to a wall or the ceiling. Many access points come with mounting hardware designed to hang the access point from a suspended ceiling.

The most important part of installing the access point is choosing the location. Ideally, it should be centrally located within the space it needs to provide access to. And it will need to have access to both power and the wired network. That's why it's important that you plan for wireless installation when you create the network's cabling plan.

If you don't have an electrical outlet at the location where you want to place the access point, you can use a Power over Ethernet (PoE) switch, which injects power on the Ethernet cable. That way, the power and network access are delivered to the access point over a single cable. Of course, you'll need to make sure that your access point is compatible with PoE to use this solution.

An alternative to using a PoE switch is to use a device called a *PoE injector*. A PoE injector has two Ethernet jacks — an input and an output — plus a power cable. The input jack accepts a standard Ethernet cable coming from a switch. You can then connect one end of an Ethernet cable to the injector's output jack and the other end to the access point. The PoE injector adds power to the output jack to provide power for the access point.

Configuring a Wireless Access Point

The software configuration for an access point is a little more involved but still not very complicated. It's usually done via a web interface. To get to the configuration page for the access point, you need to know the access point's Internet Protocol (IP) address. Then you just type that address in the address bar of a browser on any computer on the network. If you aren't certain of the IP address, you can usually find out by checking your Dynamic Host Configuration Protocol (DHCP) server.

Figure 2-1 shows the main Wireless Settings page for a typical WAP. I called up this configuration page by entering the IP address of the access point (10.0.0.145) in the address bar of a web browser and then supplying the login credentials when prompted.

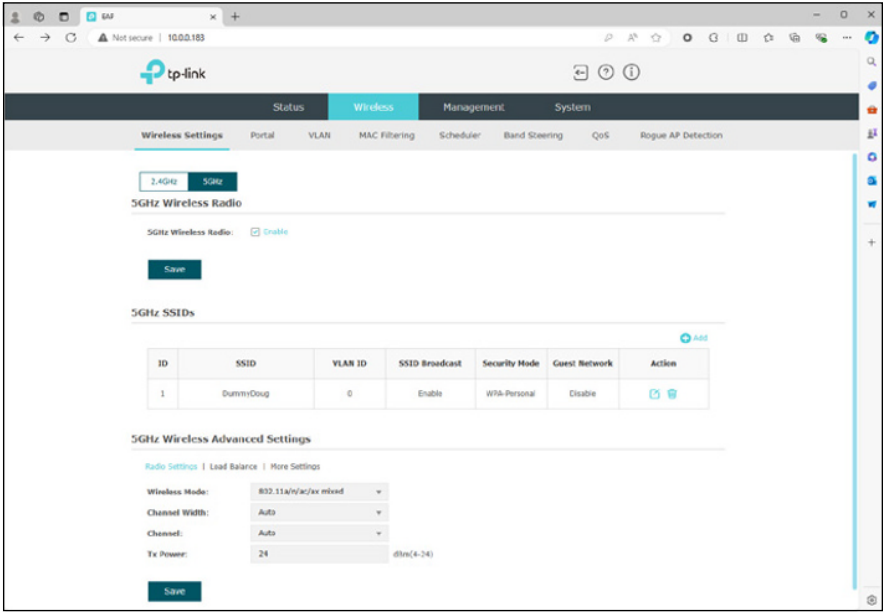


FIGURE 2-1:
The Wireless
Settings page for
a typical wireless
router.

When configuring a WAP, you’ll want to consider the following options:

- » **Broadcast frequency:** Most APs support wireless communication on two frequencies: 2.4 GHz and 5 GHz. 5 GHz wireless networks are faster than 2.4 GHz networks but have a shorter range. In most cases, you’ll want to enable both frequencies.
- » **SSID:** The SSID is the name of the wireless network that your users will recognize when they connect wirelessly. Choose a name that identifies your network and distinguishes it from wireless networks that originate from nearby businesses or homes. (*SSID* stands for Service Set Identifier, but that won’t be on the test.)
- » **VLAN:** By default, any new SSID you create will be associated with VLAN 0. If you want, you can associate an SSID with a different VLAN by enabling VLAN for the SSID and specifying a VLANID. For more information about virtual local area networks (VLANs), refer to Book 1, Chapter 3.

» **Security:** Always enable the strongest possible form of security for every one of your SSIDs. The only exception is if you enable a guest wireless network, which will allow visitors to connect to your wireless network for internet access only, with no access to any other resources connected to your local area network (LAN).

Connecting to a Wireless Network

Connecting to a wireless network on a Windows computer is straightforward. Windows automatically detects any wireless networks that are in range and displays them in a list when you tap the Wireless icon at the bottom of the screen, as shown in Figure 2-2.

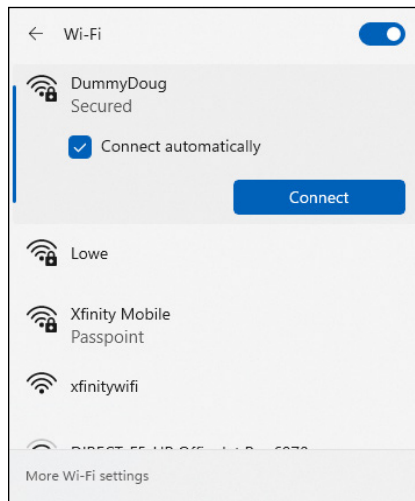


FIGURE 2-2:
Choosing a wireless network in Windows 11.

To connect to a network, just tap it, and then enter the security key when prompted. If the key is correct, you'll be connected.

At the time you connect, you can choose to connect to the network automatically whenever it's in range. If you select this option, you won't have to select the network manually or enter the security key; you'll just be connected automatically.

Windows remembers every network you connect to, which is a plus for networks you frequently use but a drawback for networks you'll likely never use again. To tell Windows to forget a network, follow these steps:

1. **Click Start, and then tap Settings.**

The Settings window appears.

2. **Click Network & Internet.**

This brings up the Network & Internet page.

3. **Click Wi-Fi.**

This brings up the Wi-Fi settings page.

4. **Click Manage Known Networks.**

This brings up the Manage Known Networks page, shown in Figure 2-3.

5. **In the Manage Known Networks section, click the Forget button for the network you want to forget.**

The network will be forgotten. To log into this network again, you'll have to enter the security key.

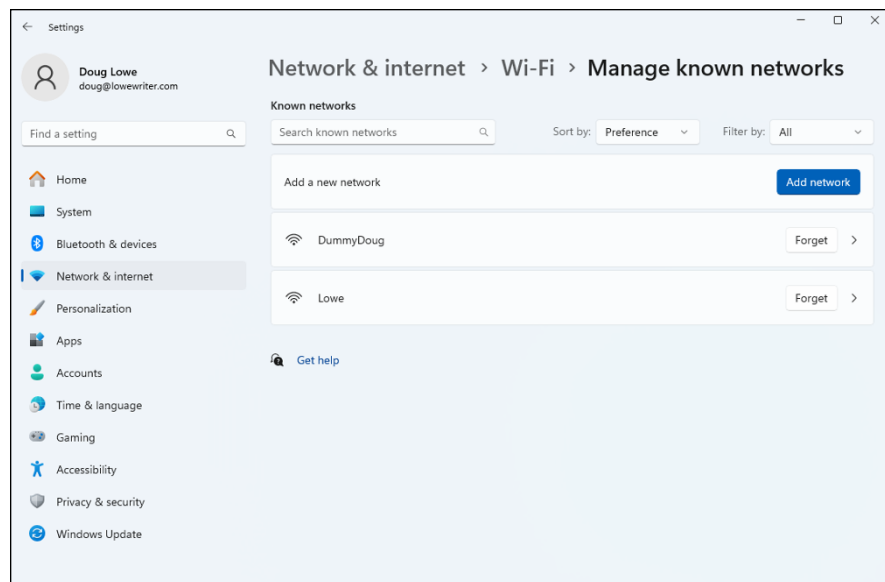


FIGURE 2-3:
Forgetting a
wireless network
in Windows 11.

Paying Attention to Wireless Network Security

Before you dive headfirst into the deep end of the wireless networking pool, you should consider the inherent security risks in setting up a wireless network. With a cabled network, the best security tool that you have is the lock on the front door

of your office. Unless someone can physically get to one of the computers on your network, they can't get into your network.

If you go wireless, an intruder doesn't have to get into your office to hack into your network. They can do it from the office next door. Or the lobby. Or the parking garage below your office. Or the sidewalk outside. In short, when you introduce wireless devices into your network, you usher in a whole new set of security issues to deal with.

Understanding wireless security threats

Wireless networks have the same basic security considerations as wired networks. As a network administrator, you need to balance the need of legitimate users to access network resources against the risk of illegitimate users breaking into your network. That's the basic dilemma of network security. Whether the network uses cables, wireless devices, kite strings and tin cans, or smoke signals, the basic issues are the same.

At one extreme of the wireless network security spectrum is the totally open network, in which anyone within range of your wireless transmissions can log on as an administrator and gain full access to every detail of your network. At the other end is what I call the "cone-of-silence syndrome," in which the network is so secure that no one can gain access to the network — not even legitimate users.

The goal of securing a wireless network is to find the happy medium between these two extremes that meets the access and risk-management needs of your organization.

The following sections describe the types of security threats that wireless networks are most likely to encounter. You should take each of these kinds of threats into consideration when you plan your network's security.

Intruders

With a wired network, an intruder usually must gain access to your facility to physically connect to your network. That's not so with a wireless network. In fact, hackers equipped with notebooks that have wireless network capability can gain access to your network if they can place themselves physically within range of your network's radio signals. Consider these possibilities:

- » If you share a building with other tenants, the other tenants' offices may be within range.
- » If you're in a multifloor building, the floor immediately above or below you may be in range.

- » The lobby outside your office may be within range of your network.
- » The parking lot outside or the parking garage in the basement may be in range.

If a would-be intruder can't get within normal broadcast range, they may try one of several tricks to increase the range:

- » A would-be intruder can switch to a bigger antenna to extend the range of their wireless computer. Some experiments have shown that big antennas can receive signals from wireless networks miles away. In fact, I once read about someone who listened in on wireless networks based in San Francisco from the Berkeley hills, across San Francisco Bay.
- » If a would-be intruder is serious about breaking into your network, they may smuggle a wireless repeater device into your facility — or near it — to extend the range of your wireless network to a location that they *can* get to.



REMEMBER

A *physical* connection to your network isn't the only way an intruder can gain access, of course. You must still take steps to prevent an intruder from sneaking into your network through your internet gateway. In most cases, this means that you need to set up a firewall to block unwanted and unauthorized traffic.

Freeloaders

Freeloaders are intruders who want to piggyback on your wireless network to get free access to the internet. If they manage to gain access to your wireless network, they probably won't do anything malicious: They'll just fire up their web browsers and surf the internet. These are folks who are too cheap to spend \$40 per month on their own broadband connection at home, so they'd rather drive into your parking lot and steal yours.

Even though freeloaders may be relatively benign, they can be a potential source of trouble. In particular:

- » Freeloaders use bandwidth that you're paying for. As a result, their mere presence can slow down internet access for your legitimate users.
- » After freeloaders gain internet access through your network, they can potentially cause trouble for you or your organization. They may use your network to download illegal pornography, or they may try to send spam via your mail server. Most ISPs will cut you off cold if they catch you sending spam, and they won't believe you when you tell them that the spam came from a kid parked in a Pinto out in your parking lot.
- » If you're in the business of *selling* access to your wireless network, obviously, freeloaders are a problem.

- » Freeloaders may start out innocently looking for free internet access. Once they get in, though, curiosity may get the better of them, leading them to snoop around your network.
- » If freeloaders can get in, so can malicious intruders.

Eavesdroppers

Eavesdroppers just like to listen to your network traffic. They don't actually try to gain access via your wireless network — at least, not at first. They just listen.

Unfortunately, wireless networks give them plenty to listen to:

- » Most WAPs regularly broadcast their Service Set Identifiers (SSIDs) to anyone who's listening.
- » When a legitimate wireless network user joins the network, an exchange of packets occurs as the network authenticates the user. An eavesdropper can capture these packets and, if security isn't set up right, determine the user's logon name and password.
- » An eavesdropper can steal files that are opened from a network server. If a wireless user opens a confidential sales report that's saved on the network, the sales-report document is broken into packets that are sent over the wireless network to the user. A skilled eavesdropper can copy those packets and reconstruct the file.
- » When a wireless user connects to the internet, an eavesdropper can see any packets that the user sends to or receives from the internet. If the user purchases something online, the transaction may include a credit card number and other personal information. (Ideally, these packets will be encrypted so that the eavesdropper won't be able to decipher the data.)

Spoilers

A *spoiler* is a hacker who gets kicks from jamming networks so that they become unusable. A spoiler usually accomplishes this act by flooding the network with meaningless traffic so that legitimate traffic gets lost in the flow. Spoilers may also try to place viruses or worm programs on your network via an unsecured wireless connection.

Rogue access points

One of the biggest problems that network administrators have to deal with is the problem of rogue access points. A *rogue access point* is an access point that

suddenly appears on your network out of nowhere. What usually happens is that an employee decides to connect a notebook computer to the network via a wireless computer. So this user stops at Computers-R-Us on the way home from work one day, buys a Fisher-Price WAP for \$25, and plugs it into the network without asking permission.

Now, in spite of all the elaborate security precautions you've taken to fence in your network, this well-meaning user has opened the barn door. It's *very* unlikely that the user will enable the security features of the WAP; in fact, they probably isn't even aware that wireless access devices *have* security features.

Unless you take some kind of action to find it, a rogue access point can operate undetected on your network for months or even years. You may not discover it until you report to work one day and find that your network has been trashed by an intruder who found their way into your network via an unprotected WAP that you didn't even know existed.



TIP

Here are some steps you can take to reduce the risk of rogue access points appearing on your system:

- » Establish a policy prohibiting users from installing WAPs on their own. Then make sure that you inform all network users of the policy, and let them know why installing an access point on their own can be such a major problem.
- » If possible, establish a program that quickly and inexpensively grants wireless access to users who want it. Rogue access points show up in the first place for two reasons:
 - Users need the access.
 - The access is hard to get through existing channels.
- » If you make it easier for users to get legitimate wireless access, you're less likely to find WAPs hidden behind file cabinets or in flower pots.
- » Once in a while, take a walk through the premises, looking for rogue access points. Take a look at every network outlet in the building; see what's connected to it.
- » Turn off all your WAPs and then walk around the premises with a wireless-equipped mobile device such as a smartphone and look for wireless networks that pop up. Just because you detect a wireless network, of course, doesn't mean you've found a rogue access point; you may have stumbled onto a wireless network in a nearby office or home. But knowing what wireless networks are available from within your office will help you determine whether or not any rogue access points exist.

Securing your wireless network

I hope you're convinced that wireless networks do, indeed, pose many security risks. In the following sections, I describe some steps that you can take to help secure your wireless network.

Changing the password

Probably the first thing you should do when you install a WAP is change its administrative password. Most access points have a built-in, web-based setup page that you can access from any web browser to configure the access point's settings. The setup page is protected by a username and password, but the username and password are initially set to default values that are easy to guess.

For Linksys access points, for example, the default username is usually either blank or `admin`, and the password is `admin`. If you leave the username and password set to their default values, anyone can access the access point and change its configuration settings, thus bypassing any other security features that you enable for the access point.

So, the first step in securing your WAP is changing the setup password to a value that can't be guessed. I suggest that you use a random combination of numerals and both uppercase and lowercase letters. Be sure to store the password in a secure location. (If you forget the password, you can press the Reset button on the router to restore it to its factory default. Then you can log on by using the default password, which you can find with the documentation that came with the router.)

Securing the SSID

The next step is to secure the SSID that identifies the network. A client must know the access point's SSID to join the wireless network. If you can prevent unauthorized clients from discovering the SSID, you can prevent them from accessing your network.



WARNING

Securing the SSID isn't a complete security solution, so you shouldn't rely on it as your only security mechanism. SSID security can slow down casual intruders who are just looking for easy and free internet access, but it isn't possible to prevent serious hackers from discovering your SSID.



TIP

You can do three things to secure your SSID:

- » **Change the SSID from the default.** Most access points come preconfigured with well-known default SSIDs, such as those listed in Table 2-1. By changing your access point's SSID, you can make it more difficult for an intruder to determine your SSID and gain access.



WARNING

» **Disable SSID broadcast.** Most access points frequently broadcast their SSIDs so that clients can discover the network when they come within range. Clients that receive this SSID broadcast can use the SSID to join the network.

You can increase network security somewhat by disabling the SSID broadcast feature. That way, clients won't automatically learn the access point's SSID. To join the network, a client computer must figure out the SSID on its own. Then you can tell your wireless network users the SSID to use when they configure their clients.

Unfortunately, when a client computer connects to a wireless network, it sends the SSID to the access point in an unencrypted packet. So a sophisticated intruder who's using a packet sniffer to eavesdrop on your wireless network can determine your SSID as soon as any legitimate computer joins the network.

» **Disable guest mode.** Many access points have a guest-mode feature that enables client computers to specify a blank SSID or to specify "any" as the SSID. If you want to ensure that only clients that know the SSID can join the network, you must disable this feature.

TABLE 2-1

Common Default SSID Values

SSID	Manufacturer
3com	3Com
Compaq	Compaq
Linksys	Linksys
Tsunami	Cisco
Wireless	NetGear
WLAN	D-Link
WLAN	SMC

Using WPA and WPA2

WPA, which stands for *Wi-Fi Protected Access*, is a security protocol for wireless networks. You should always ensure that WPA security is enabled. Figure 2-4 shows a typical configuration screen for a WAP that confirms that WPA has been enabled on the device.

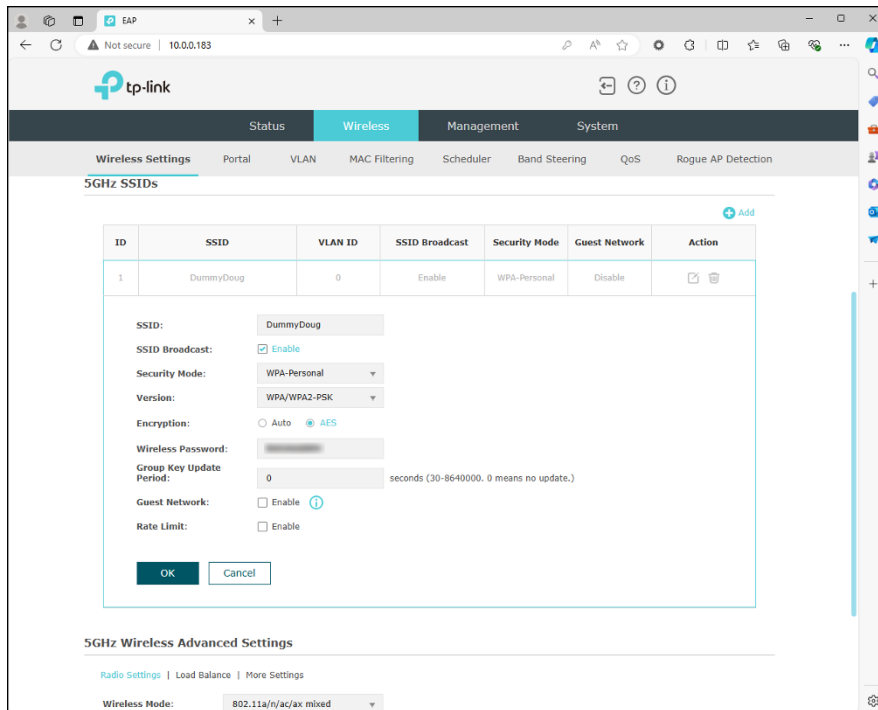


FIGURE 2-4:
Changing the
WPA settings on a
wireless router.

Here are a few additional things to know about WPA:

- » A small office and home version of WPA, called WPA-PSK, bases its encryption keys on a passkey value that you supply. True WPA devices, however, rely on a special authentication server to generate the keys.
- » All versions of Windows since Windows XP Service Pack 2 have built-in support for WPA.
- » The official IEEE standard for WPA is 802.11i. However, WPA devices were widely available before the 802.11i standard was finalized; as a result, not all WPA devices implement every aspect of 802.11i.
- » The original version of WPA has been superseded by a newer version, named WPA2.

Using MAC address filtering

MAC address filtering allows you to specify a list of MAC addresses for the devices that are allowed to access the network or are prohibited from accessing the network. If a computer with a different MAC address tries to join the network via the access point, the access point will deny access.

MAC address filtering is a great idea for wireless networks with a fixed number of clients. If you set up a wireless network at your office so that a few workers can connect their notebook computers, you can specify the MAC addresses of those computers in the MAC filtering table. Then other computers won't be able to access the network via the access point.



WARNING

Unfortunately, it isn't difficult to configure a computer to lie about its MAC address. Thus, after a potential intruder determines that MAC filtering is being used, they can just sniff packets to determine an authorized MAC address and then configure their computer to use that address. (This practice is called *MAC spoofing*.) So you shouldn't rely on MAC address filtering as your only means of security.

Figure 2-5 shows the screen used to edit the MAC address table for a typical WAP.

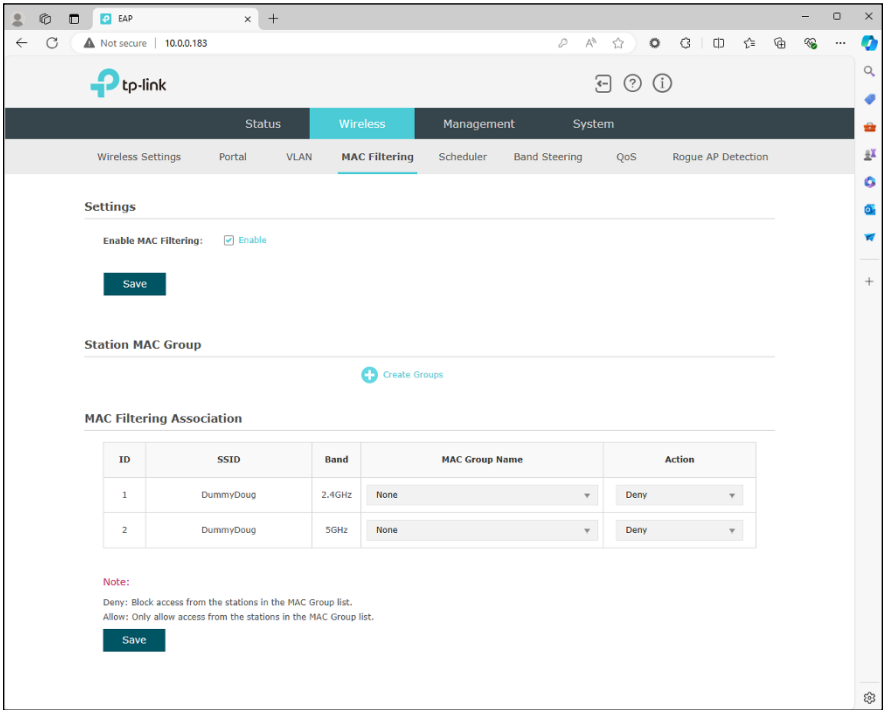


FIGURE 2-5:
A MAC address table for a wireless router.

Placing your access points outside the firewall

The most effective security technique for wireless networking is placing all your WAPs *outside* your firewall. That way, all network traffic from wireless users will have to travel through the firewall to access the network.

DON'T NEGLECT THE BASICS

The security techniques described in this chapter are specific to wireless networks. They should be used alongside the basic security techniques that are presented in Book 1, Chapter 4 and in Book 10. In other words, don't forget the basics, such as these:

- Use strong passwords for your user accounts.
- Apply security patches to your servers.
- Change default server account information (especially the administrator password).
- Disable unnecessary services.
- Check your server logs regularly.
- Install virus protection.
- Back up!

As you can imagine, doing this can significantly limit network access for wireless users. To get around those limitations, you can enable a virtual private network (VPN) connection for your wireless users. The VPN will allow full network access to authorized wireless users.

Obviously, this solution requires a bit of work to set up and can be a little inconvenient for your users, but it's an excellent way to fully secure your WAPs.

Troubleshooting a wireless network

Wireless networks are great until something goes haywire. When a regular network doesn't work, you usually know about it right away because the network simply becomes unavailable. You can't display web pages, read email, or access files on shared drives.

The troubleshooting chapters in Book 9, Chapter 4 address the most common problems encountered on cabled networks. But wireless networks can cause problems of their own. And to add to the frustration, wireless networks tend to degrade rather than completely fail. Performance gets slower. Web pages that usually pop up in a second or two take 15 to 20 seconds to appear. Or sometimes they don't appear at all, but if you try again a few minutes later, they download fine.

This following sections offer some troubleshooting tips that can help you restore normalcy to a failing wireless network.

Checking for obvious problems

Before you roll up your sleeves and take drastic corrective action, you should check for a few obvious things if you're having wireless network trouble. The following list highlights some basic things you should check for:

- » Is everything turned on? Make sure you have lights on your WAP/router, as well as on your cable or DSL modem.
- » Many access point/routers use a power supply transformer that plugs into the wall. Make sure that the transformer is plugged into the wall outlet and that the small cable that comes out of the transformer is plugged into the power connector on the access point/router.
- » Are the cables connected? Check the network cable that connects your access point/router to the cable or DSL modem.
- » Try restarting everything. Turn off the computer, the access point/router, and your cable or DSL modem. Leave everything off for at least two minutes. Then turn everything back on. Sometimes, simply cycling the power off and back on clears up a connection problem.

Pinpointing the problem

If you can't connect to the internet, one of the first steps (after you've made sure that everything is turned on) is finding out whether the problem is with your access point/router or with your broadband connection. Here is one way you can check to find out whether your wireless connection is working:

1. **Open a command prompt window by choosing Start→cmd, and pressing Enter.**
2. **At the command prompt, type ipconfig, and press Enter.**

You should get a display similar to this:

```
Ethernet adapter Wireless Network Connection:
Connection-specific DNS Suffix . : hsd1.ca.comcast.net)).
IP Address . . . . . : 10.0.0.200
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.0.0.1
```

If the display resembles this but with different numbers, you're connected to the wireless network, and the problem most likely lies with your broadband modem.

But if the IP Address, Subnet Mask, and Default Gateway indicate 0.0.0.0 instead of valid IP addresses, you have a problem with your wireless network.

Changing channels

One of the most common sources of wireless network trouble is interference from other wireless devices. The culprit might be a cordless phone, or it could be a neighbor who also has a wireless network.

The simplest solution to this type of interference is changing channels. 802.11b access points let you select 1 of 11 different channels to broadcast on. If you're having trouble connecting to your access point, try changing the channel. To do that, you must log on to the router with the administrator password. Then hunt around the router's administrator pages until you find the controls that let you change the channel.

You may have to try changing the channel several times before you solve the problem. Unfortunately, 802.11b channels overlap slightly, which means that broadcasts on one channel may interfere with broadcasts on adjacent channels. Thus, if you're having trouble connecting on channel 1, don't bother switching to channel 2. Instead, try switching to channel 5 or 6. If that doesn't work, switch to channel 10 or 11.

Fiddling with the antennas

Sometimes, you can fix intermittent connection problems by fiddling with the antennas on the access point and your computer's wireless adapter. This procedure is similar to playing with old-fashioned rabbit-ear antennas on a TV to get the best reception.

The angles of the antennas sometimes make a difference, so try adjusting the antenna angles. In addition, you usually have better results if you place the access point at a high location, such as on top of a bookshelf.

In some cases, you may actually need to add a high-gain antenna to the access point to increase its range. A high-gain antenna simply snaps or screws onto the access point to provide a bigger antenna. Antennas such as these cost about \$70 for a pair.

A more drastic fix is to add a signal booster to your access point. A *signal booster* is a power amplifier that increases the transmission power of most wireless devices by a factor of five. A typical signal booster costs about \$100.

Adding another access point

If you have a computer that's out of range of your access point, one solution is to add a second access point closer to the problematic computer. Most likely, the

only difficulty will be getting an Ethernet cable to the location where you want to put your second access point.

If possible, you can simply run a length of cable through your walls or attic to the second access point. If that solution isn't feasible, you can use a HomePlug or HomePNA network connection for the second access point.

An alternative to a second access point is simply adding a range extender. This handy device plugs directly into any electrical outlet and provides a pass-through outlet so you can still use your vacuum cleaner. Just plug this device midway between your access point and the computer that's having trouble connecting. (Note that using a range extender will slow down your Wi-Fi connection.)

Help! I forgot my router's password!

I mention many times throughout this book that you should always change the default passwords that come with computer and operating systems to more secure passwords, usually consisting of a random combination of letters, digits, and special symbols.

Ideally, you've already taken my sage advice and changed the password on your combination WAP/router. Good for you. But what if you forget the password later? Is there any way to get back into your access point/router then?



TIP

Fortunately, there is. Most access point/routers have a Reset button. It's usually located on the back or on the bottom of the router's case. Press this button to restore the access point/router to its factory default settings. That action resets the administrator password to the factory default — and also resets any other custom settings you've applied, so you may have to reconfigure your router to get it working again.

